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- ✓ Course framework
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AP[®] Physics C: Electricity and Magnetism

COURSE AND EXAM DESCRIPTION

Effective
Fall 2024

ADVANCED PLACEMENT PHYSICS C: ELECTRICITY AND MAGNETISM

TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS	
Coulomb constant,	$k = \frac{1}{4\pi\epsilon_0} = 9.0 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2}$
Vacuum permittivity,	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 / (\text{N} \cdot \text{m}^2)$
Vacuum permeability,	$\mu_0 = 4\pi \times 10^{-7} (\text{T} \cdot \text{m}) / \text{A}$
Proton mass,	$m_p = 1.67 \times 10^{-27} \text{ kg}$
Neutron mass,	$m_n = 1.67 \times 10^{-27} \text{ kg}$
Electron mass,	$m_e = 9.11 \times 10^{-31} \text{ kg}$
Elementary charge,	$e = 1.60 \times 10^{-19} \text{ C}$
1 electron volt,	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
Speed of light,	$c = 3.00 \times 10^8 \text{ m/s}$
1 unified atomic mass unit,	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg} = 931 \text{ MeV}/c^2$
Universal gravitational constant, $G = 6.67 \times 10^{-11} \text{ m}^3 / (\text{kg} \cdot \text{s}^2) = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2$	
Magnitude of the acceleration due to gravity at Earth's surface, $g = 9.8 \text{ m/s}^2$	
Magnitude of the gravitational field strength at Earth's surface, $g = 9.8 \text{ N/kg}$	

UNIT SYMBOLS	
ampere,	A
coulomb,	C
electron volt,	eV
farad,	F
henry,	H
hertz,	Hz
joule,	J
kilogram,	kg
meter,	m
newton,	N
ohm,	Ω
second,	s
tesla,	T
volt,	V
watt,	W

PREFIXES		
Factor	Prefix	Symbol
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES							
θ	0°	30°	37°	45°	53°	60°	90°
$\sin \theta$	0	1/2	3/5	$\sqrt{2}/2$	4/5	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	4/5	$\sqrt{2}/2$	3/5	1/2	0
$\tan \theta$	0	$\sqrt{3}/3$	3/4	1	4/3	$\sqrt{3}$	∞

The following conventions are used in this exam:

- The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- Air resistance is assumed to be negligible unless otherwise stated.
- Springs and strings are assumed to be ideal unless otherwise stated.
- The electric potential is zero at an infinite distance from an isolated point charge.
- The direction of current is the direction in which positive charges would drift.
- All batteries, wires, and meters are assumed to be ideal unless otherwise stated.

ELECTRICITY AND MAGNETISM

$$|\vec{F}_E| = \frac{1}{4\pi\epsilon_0} \frac{|q_1 q_2|}{r^2} = k \frac{|q_1 q_2|}{r^2}$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{r}$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$Q_{\text{total}} = \int \rho(r) dV$$

$$U_E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$

$$\Delta V = - \int_a^b \vec{E} \cdot d\vec{r}$$

$$E_x = - \frac{dV}{dx}$$

$$\Delta U_E = q\Delta V$$

$$C = \frac{Q}{\Delta V}$$

$$C = \frac{\kappa\epsilon_0 A}{d}$$

$$U_C = \frac{1}{2} Q\Delta V$$

$$\kappa = \frac{\epsilon}{\epsilon_0}$$

$$I = \frac{dq}{dt}$$

$$I = \int \vec{J} \cdot d\vec{A}$$

$$\vec{E} = \rho\vec{J}$$

$$R = \frac{\rho\ell}{A}$$

$$I = \frac{\Delta V}{R}$$

$$P = I\Delta V$$

A = area

C = capacitance

d = distance

E = electric field

F = force

I = current

J = current density

ℓ = length

P = power

q = charge

Q = charge

r = radius, distance, or position

R = resistance

t = time

U = potential energy

V = electric potential or volume

ϵ = electric permittivity

ρ = resistivity or charge density

κ = dielectric constant

Φ = flux

$$R_{\text{eq},s} = \sum_i R_i$$

$$\frac{1}{R_{\text{eq},p}} = \sum_i \frac{1}{R_i}$$

$$\frac{1}{C_{\text{eq},s}} = \sum_i \frac{1}{C_i}$$

$$C_{\text{eq},p} = \sum_i C_i$$

$$\tau = R_{\text{eq}} C_{\text{eq}}$$

$$\oint \vec{B} \cdot d\vec{A} = 0$$

$$\vec{F}_B = q(\vec{v} \times \vec{B})$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I(d\vec{\ell} \times \hat{r})}{r^2}$$

$$\vec{F}_B = \int I(d\vec{\ell} \times \vec{B})$$

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}}$$

$$B_{\text{sol}} = \mu_0 nI$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{\ell} = - \frac{d\Phi_B}{dt}$$

$$|\mathcal{E}_{\text{sol}}| = N \left| \frac{d\Phi_B}{dt} \right|$$

$$L_{\text{sol}} = \frac{\mu_{\text{core}} N^2 A}{\ell}$$

$$U_L = \frac{1}{2} LI^2$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$\tau = \frac{L}{R_{\text{eq}}}$$

$$\omega_{LC} = \frac{1}{\sqrt{LC}}$$

A = area

B = magnetic field

C = capacitance

F = force

I = current

ℓ = length

L = inductance

n = number of loops per unit length

N = number of loops

q = charge

r = radius, distance, or position

R = resistance

t = time

U = potential energy

v = velocity or speed

$\mathcal{E}$ = emf

μ = magnetic permeability

τ = time constant

Φ = flux

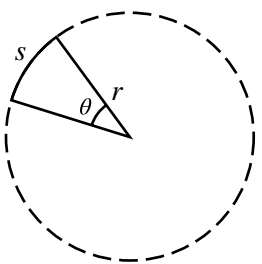
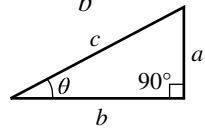
ω = angular frequency

MECHANICS

$v_x = v_{x0} + a_x t$	a = acceleration
$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	E = energy
$v_x^2 = v_{x0}^2 + 2a_x (x - x_0)$	f = frequency
$\Delta x = \int v_x(t) dt$	F = force
$\Delta v_x = \int a_x(t) dt$	h = height
$\vec{x}_{\text{cm}} = \frac{\sum m_i \vec{x}_i}{\sum m_i}$	J = impulse
$\vec{r}_{\text{cm}} = \frac{\int \vec{r} dm}{\int dm}$	k = spring constant
$\lambda = \frac{d}{d\ell} m(\ell)$	K = kinetic energy
$\vec{a}_{\text{sys}} = \frac{\sum \vec{F}}{m_{\text{sys}}} = \frac{\vec{F}_{\text{net}}}{m_{\text{sys}}}$	ℓ = length
$ \vec{F}_g = G \frac{m_1 m_2}{r^2}$	m = mass
$ \vec{F}_f \leq \mu \vec{F}_N $	M = mass
$\vec{F}_s = -k \Delta \vec{x}$	p = momentum
$a_c = \frac{v^2}{r} = r \omega^2$	P = power
$T = \frac{1}{f}$	r = radius, distance, or position
$K = \frac{1}{2} m v^2$	t = time
$W = \int_a^b \vec{F} \cdot d\vec{r}$	T = period
$\Delta K = \sum W_i = \sum F_{\parallel,i} d_i$	U = potential energy
$\Delta U = - \int_a^b \vec{F}_{\text{cf}}(r) \cdot d\vec{r}$	v = velocity or speed
$F_x = - \frac{dU(x)}{dx}$	W = work
$U_s = \frac{1}{2} k (\Delta x)^2$	x = position or distance
$U_G = -G \frac{m_1 m_2}{r}$	y = height
$\Delta U_g = mg \Delta y$	λ = linear mass density
	μ = coefficient of friction

$\omega = \frac{d\theta}{dt}$	a = acceleration
$\alpha = \frac{d\omega}{dt}$	d = distance
$\omega = \omega_0 + \alpha t$	f = frequency
$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$	F = force
$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$	I = rotational inertia
$v = r \omega$	k = spring constant
$a_T = r \alpha$	K = kinetic energy
$\vec{\tau} = \vec{r} \times \vec{F}$	ℓ = length
$I_{\text{tot}} = \sum I_i = \sum m_i r_i^2$	L = angular momentum
$I = \int r^2 dm$	m = mass
$I' = I_{\text{cm}} + M d^2$	M = mass
$\alpha_{\text{sys}} = \frac{\sum \tau}{I_{\text{sys}}} = \frac{\tau_{\text{net}}}{I_{\text{sys}}}$	p = momentum
$K_{\text{rot}} = \frac{1}{2} I \omega^2$	r = radius, distance, or position
$W = \int \tau \cdot d\theta$	t = time
$\vec{L} = \vec{r} \times \vec{p} = I \vec{\omega}$	T = period
$\Delta L = \int \tau dt$	v = velocity or speed
$\Delta x_{\text{cm}} = r \Delta \theta$	W = work
$T = \frac{2\pi}{\omega} = \frac{1}{f}$	x = position or distance
$T_s = 2\pi \sqrt{\frac{m}{k}}$	α = angular acceleration
$T_p = 2\pi \sqrt{\frac{\ell}{g}}$	θ = angle
$T_{\text{phys}} = 2\pi \sqrt{\frac{I}{mgd}}$	τ = torque
$x = x_{\text{max}} \cos(\omega t + \phi)$	ϕ = phase angle
	ω = angular frequency or angular speed

GEOMETRY AND TRIGONOMETRY

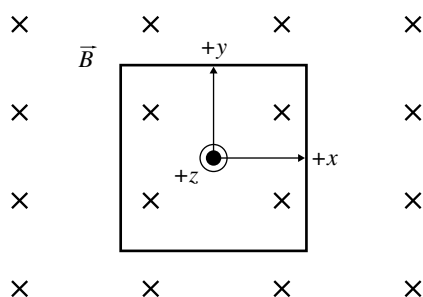
Rectangle $A = bh$	Rectangular Solid $V = \ell wh$		$A = \text{area}$ $b = \text{base}$ $C = \text{circumference}$ $h = \text{height}$ $\ell = \text{length}$ $r = \text{radius}$ $s = \text{arc length}$ $S = \text{surface area}$ $V = \text{volume}$ $w = \text{width}$ $\theta = \text{angle}$	Right Triangle $a^2 + b^2 = c^2$ $\sin \theta = \frac{a}{c}$ $\cos \theta = \frac{b}{c}$ $\tan \theta = \frac{a}{b}$
Triangle $A = \frac{1}{2}bh$	Cylinder $V = \pi r^2 \ell$ $S = 2\pi r \ell + 2\pi r^2$			
Circle $A = \pi r^2$ $C = 2\pi r$ $s = r\theta$	Sphere $V = \frac{4}{3}\pi r^3$ $S = 4\pi r^2$			

VECTORS	CALCULUS	IDENTITIES
$\vec{A} \cdot \vec{B} = AB \cos \theta$ $ \vec{A} \times \vec{B} = AB \sin \theta$ $\vec{r} = (A\hat{i} + B\hat{j} + C\hat{k})$ $\vec{C} = \vec{A} + \vec{B}$ $\vec{C} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j}$	$\frac{df}{dx} = \frac{df}{du} \frac{du}{dx}$ $\frac{d}{dx}(x^n) = nx^{n-1}$ $\frac{d}{dx}(e^{ax}) = ae^{ax}$ $\frac{d}{dx}(\ln ax) = \frac{1}{x}$ $\frac{d}{dx}[\sin(ax)] = a \cos(ax)$ $\frac{d}{dx}[\cos(ax)] = -a \sin(ax)$ $\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1$ $\int e^{ax} dx = \frac{1}{a} e^{ax}$ $\int \frac{dx}{x+a} = \ln x+a $ $\int \cos(ax) dx = \frac{1}{a} \sin(ax)$ $\int \sin(ax) dx = -\frac{1}{a} \cos(ax)$	$\log(a \cdot b^x) = \log a + x \log b$ $\sin^2 \theta + \cos^2 \theta = 1$ $\sin(2\theta) = 2 \sin \theta \cos \theta$ $\frac{\sin \theta}{\cos \theta} = \tan \theta$

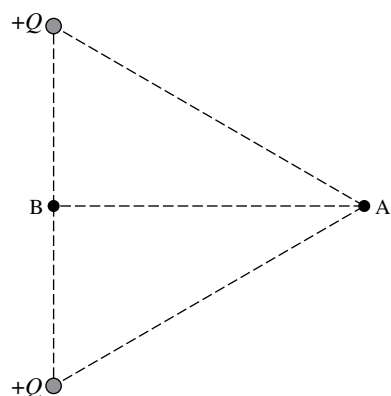
Sample Exam Questions

The sample exam questions that follow illustrate the relationship between the course framework and the AP Physics C: Electricity and Magnetism Exam and serve as examples of the types of questions that appear on the exam. These sample questions do not represent the full range and distribution of items on an official AP Physics C: Electricity and Magnetism Exam. After the sample questions is a table which shows which skill, learning objective, and essential knowledge statement each question relates to. This table also provides the answers to the multiple-choice questions.

Section I: Multiple-Choice Questions



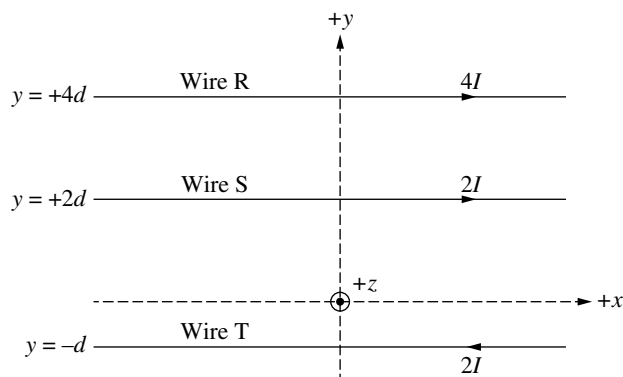
1. A rectangular loop of wire lies in the xy -plane in a region with a uniform magnetic field $\vec{B}$ directed in the $-z$ -direction, as shown. In which of the following cases will an emf be induced in the loop?
 - (A) When the loop is moving without rotating in the $+y$ -direction
 - (B) When the loop is moving without rotating in the $+z$ -direction
 - (C) When the loop is rotating about an axis along the z -axis
 - (D) When the loop is rotating about an axis along the y -axis
2. A cylindrical wire segment with length 0.25 m and diameter $3.0 \times 10^{-3}\text{ m}$ dissipates energy at a rate of $6.0 \times 10^{-4}\text{ W}$ when a current of 0.50 A is in the wire segment. Which of the following is approximately equal to the resistivity of the wire?
 - (A) $6.8 \times 10^{-8}\ \Omega \cdot \text{m}$
 - (B) $9.0 \times 10^{-5}\ \Omega \cdot \text{m}$
 - (C) $7.5 \times 10^{-3}\ \Omega \cdot \text{m}$
 - (D) $840\ \Omega \cdot \text{m}$



3. Two small spheres, each with positive charge $+Q$, are fixed in place at two corners of an equilateral triangle, as shown in the figure. Point A is at the other corner, and Point B is midway between the spheres. A test charge with positive charge $+q$ is moved from Point A to Point B at a constant velocity by an external force. The work done by the external force on the moving particle is W_{ext} , and the work done by the electrostatic forces from the spheres on the moving particle is W_{elec} . Which of the following correctly identifies the signs of these quantities?

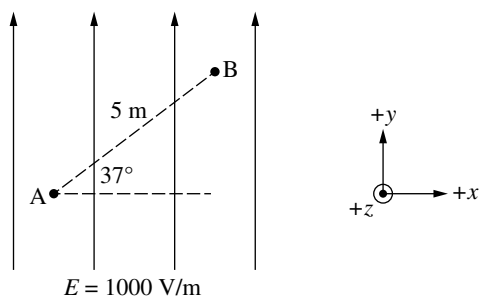
	W_{ext}	W_{elec}
A	Positive	Positive
B	Positive	Negative
C	Negative	Positive
D	Negative	Negative

- (A) A
(B) B
(C) C
(D) D



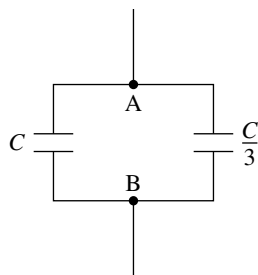
4. Three long wires, R, S, and T, are arranged in the xy -plane parallel to each other and the x -axis at the positions shown in the figure. The direction and current in each wire are indicated. Which of the following correctly indicates the magnitude and direction of the magnetic field at the origin of the coordinate system?

- (A) $\frac{\mu_0 I}{\pi d}$ in the $-z$ -direction
 (B) $\frac{\mu_0 I}{\pi d}$ in the $+z$ -direction
 (C) $\frac{2\mu_0 I}{\pi d}$ in the $-z$ -direction
 (D) $\frac{2\mu_0 I}{\pi d}$ in the $+z$ -direction

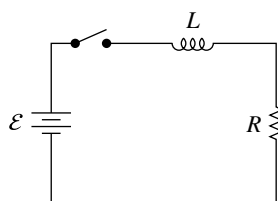


5. Points A and B are separated by a distance of 5 m and located in a uniform electric field of magnitude 1000 V/m directed in the $+y$ -direction, as shown in the figure. The line between Point A and Point B makes an angle of 37° to the x -axis. When a proton (of charge $+e$) is moved by an external force from Point A to Point B, what is the change in electric potential energy of the proton-field system?

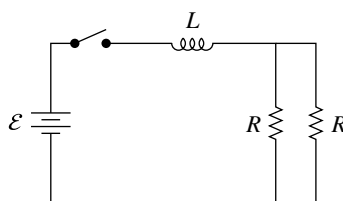
- (A) -5000 eV
 (B) -3000 eV
 (C) $+3000$ eV
 (D) $+5000$ eV



6. Two air-filled capacitors, with capacitance C and $\frac{C}{3}$, are connected in parallel, as shown. The equivalent capacitance between points A and B is C_{eq} . A dielectric is inserted into the capacitor with capacitance $\frac{C}{3}$, and the new equivalent capacitance between points A and B is $2C_{\text{eq}}$. What is the value of the dielectric constant of the dielectric?
- (A) 2
(B) 3
(C) 5
(D) 6



Circuit 1



Circuit 2

7. An inductor with inductance L is placed in series with a resistor of resistance R , a battery of emf $\mathcal{E}$, and a switch that is initially open, as shown in the Circuit 1 diagram. The switch is closed at time $t=0$, and Circuit 1 reaches half of its steady-state current at $t=t_1$. Circuit 2 is constructed by adding a resistor of resistance R in parallel with the original resistor, as shown in the Circuit 2 diagram. The switch is closed at time $t=0$, and Circuit 2 reaches half of its steady-state current at $t=t_2$. Which of the following correctly compares t_1 and t_2 , and provides a valid justification?
- (A) $t_2 = t_1$, because the time for an RL circuit to reach half of its steady-state current only depends on L
(B) $t_2 = t_1$, because both the time constant and the steady-state current double in Circuit 2 compared to Circuit 1
(C) $t_2 > t_1$, because the equivalent resistance is greater in Circuit 2 than in Circuit 1, so the time constant is greater in Circuit 2
(D) $t_2 > t_1$, because the equivalent resistance is less in Circuit 2 than in Circuit 1, so the time constant is greater in Circuit 2

Questions 8 and 9 refer to the following.

A sphere of uniform charge density has a positive net charge $+Q$ and radius R_s . A small particle of positive charge $+q$ can be moved through space both inside the sphere of uniform charge and outside the sphere.

8. The particle of charge $+q$ is initially located a distance $2R_s$ from the center of the sphere and then moved to the sphere's center. Which of the following correctly describes the change in the electric potential energy of the particle-sphere system, ΔU_{ps} , when the particle is moved, and provides a valid justification?
- (A) $\Delta U_{ps} > 0$. The electric field due to the sphere's charge does positive work on the positively charged particle.
- (B) $\Delta U_{ps} > 0$. The motion of the positively charged particle is opposite to the direction of the electric field due to the sphere's charge.
- (C) $\Delta U_{ps} = 0$. The electric field due to the sphere's charge is zero at the center of the sphere.
- (D) $\Delta U_{ps} = 0$. As the positively charged particle is moved, positive work done outside the sphere is canceled by negative work of equal magnitude that is done inside the sphere.
9. With the particle held in place at the center of the charged spherical region, what is the magnitude of the net electric field due to both the particle and the sphere at a distance $\frac{R_s}{2}$ from the sphere's center?

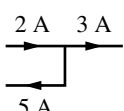
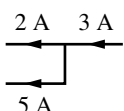
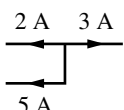
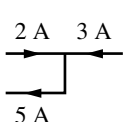
(A) $\frac{1}{4\pi\epsilon_0(R_s/2)^2} \left(q + \frac{1}{8}Q \right)$

(B) $\frac{1}{4\pi\epsilon_0(R_s/2)^2} \left(q - \frac{1}{8}Q \right)$

(C) $\frac{1}{4\pi\epsilon_0(R_s/2)^2} \left(q + \frac{1}{2}Q \right)$

(D) $\frac{1}{4\pi\epsilon_0(R_s/2)^2} \left(q - \frac{1}{2}Q \right)$

10. A junction in a circuit consists of three wires connected together. Which of the following diagrams could indicate the directions and magnitudes of the currents in the three wires?

- (A) 
- (B) 
- (C) 
- (D) 

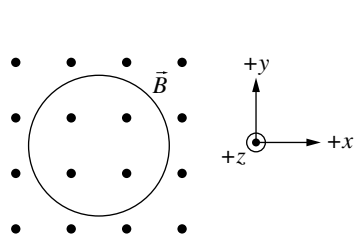


Figure 1

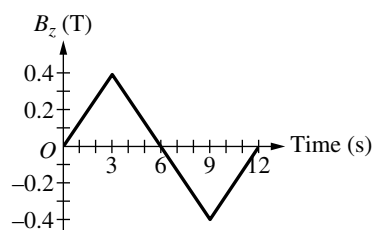
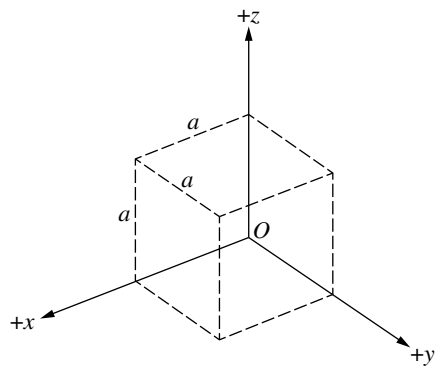


Figure 2

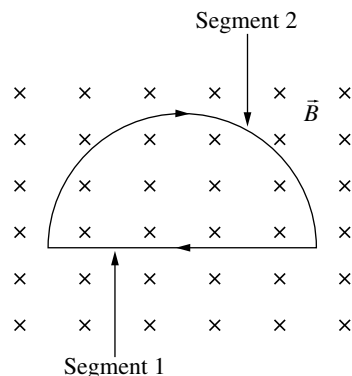
11. A metal wire of resistance $10\ \Omega$ is bent into a circular loop of radius $0.10\ \text{m}$ and placed in a uniform magnetic field $\vec{B}$, as shown in Figure 1. The z -component B_z of the magnetic field is shown as a function of time in Figure 2, where positive indicates that the magnetic field is directed out of the plane of the figure. Which of the following correctly indicates the magnitude and direction of the current induced in the loop at time $t = 6.0\ \text{s}$?

	Magnitude	Direction
A	3.8 mA	Clockwise
B	3.8 mA	Counterclockwise
C	0.42 mA	Clockwise
D	0.42 mA	Counterclockwise

- (A) A
(B) B
(C) C
(D) D



12. A Gaussian surface, in the shape of a cube of side length a , is oriented with one corner at the origin O , as shown in the figure. The cubical surface is in a region with charges (not shown) where the resulting electric field is described by $\vec{E} = (bx)\hat{i}$, where b is a positive constant. What is the total electric flux through the surface of the cube?
- (A) Zero
 - (B) ba^3
 - (C) $2ba^3$
 - (D) $6ba^3$



13. A semicircular loop with a clockwise current is placed in a uniform magnetic field that is directed into the page, as shown in the figure. $\vec{F}_1$ is the net force on Segment 1, which is the straight portion of the loop. $\vec{F}_2$ is the net force on Segment 2, which is the curved portion of the loop. Which of the following correctly indicates the directions and relative magnitudes of $\vec{F}_1$ and $\vec{F}_2$?

	Direction of $\vec{F}_1$	Direction of $\vec{F}_2$	Magnitudes
A	Toward the bottom of the figure	Toward the top of the figure	$F_1 = F_2$
B	Toward the bottom of the figure	Toward the top of the figure	$F_1 < F_2$
C	Toward the top of the figure	Toward the bottom of the figure	$F_1 = F_2$
D	Toward the top of the figure	Toward the bottom of the figure	$F_1 < F_2$

- (A) A
(B) B
(C) C
(D) D

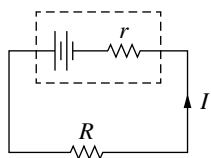


Figure 1

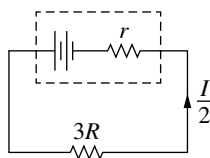
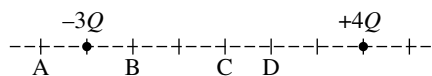


Figure 2

14. When a nonideal battery is connected to a resistor with resistance R , as shown in Figure 1, the current in the circuit is I . When the same nonideal battery is connected to a resistor with resistance $3R$, as shown in Figure 2, the current in the circuit is $\frac{I}{2}$. What is the internal resistance r of the battery?

- (A) $\frac{R}{4}$
(B) $\frac{R}{2}$
(C) R
(D) $2R$



15. Two small spheres are arranged along a line and have charges of $-3Q$ and $+4Q$, as shown in the figure. The tick marks on the axis are equally spaced. At which of the labeled tick marks does the electric field have the greatest magnitude?
- (A) A
(B) B
(C) C
(D) D

Section II: Free-Response Questions

FREE-RESPONSE QUESTION: MATHEMATICAL ROUTINES

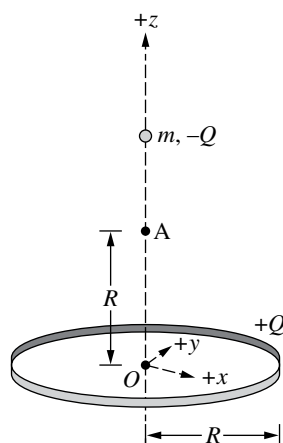


Figure 1

1. A thin, nonconducting ring is held fixed in the xy -plane with its center at the origin O . The ring has radius R and a positive charge $+Q$ uniformly distributed around its circumference. A small sphere of mass m and negative charge $-Q$ is on the z -axis, as shown in Figure 1. Point A is located on the z -axis at $z = R$.

- (a) i. In which of the following regions, if any, can the sphere be placed on the z -axis so that the net electric potential at Point A due to the sphere and the ring is zero? **Indicate** your answer by selecting one of the options below.

_____ Above Point A only _____ Below Point A only

_____ Either above or below Point A

_____ There is no location the sphere can be placed so that the net electric potential is zero

Briefly **justify** your reasoning.

- ii. The sphere is released from rest on the z -axis some distance above the ring. All forces other than the electric force exerted by the ring are negligible. The sphere passes through the origin O , where it has speed v_O , and continues moving, eventually passing through the location $z = -3R$. **Derive** an expression for the speed of the sphere at the instant the sphere passes through the location $z = -3R$. Express your answer in terms of m , Q , R , v_O , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

The sphere is then removed. Figure 2 shows the ring and Point C, which is located at $z = 2R$ on the z -axis.

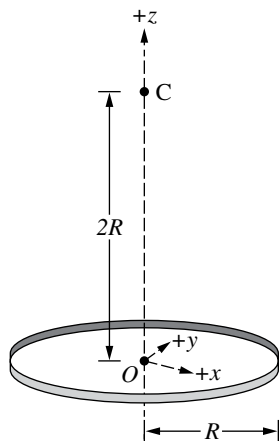


Figure 2

- (b) **Derive** an expression for the magnitude of the electric field due to the ring at Point C. Express your answer in terms of Q , R , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

FREE-RESPONSE QUESTION: TRANSLATION BETWEEN REPRESENTATIONS

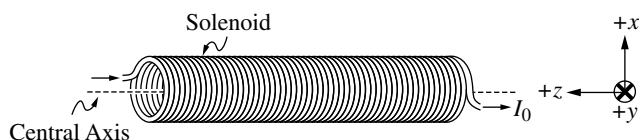


Figure 1: Side View

Note: Figure not drawn to scale

2. A long solenoid with its central axis along the z -axis has current I_0 , as shown in Figure 1. The length of the solenoid is much larger than the radius of the solenoid loops.
- (a) Figure 2 shows a cross-sectional view of the middle of the solenoid, along with three square regions labeled 1, 2, and 3. Region 1 is centered on the central axis of the solenoid, Region 2 is inside the solenoid but offset from the central axis, and Region 3 is outside the solenoid.

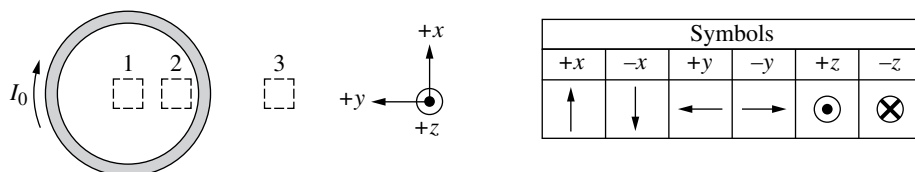


Figure 2: Cross-Sectional View

Inside the square for each region in Figure 2, **indicate** the direction of the magnetic field produced by the solenoid by drawing one of the symbols shown in the table. If the magnetic field is zero or negligible in any region, write “zero” just below that region.

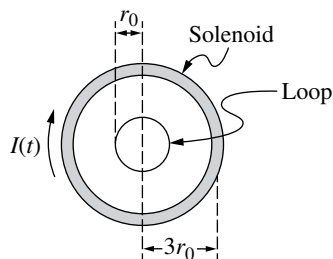


Figure 3

A small conducting loop of radius r_0 is placed at the center of the solenoid, as shown in the cross-sectional view in Figure 3. The solenoid radius is $3r_0$ and the solenoid has n_0 turns per unit of length along the axis. Starting at time $t=0$, the current I in the solenoid is varied according to the equation $I(t) = I_0(1 - bt^2)$, where I_0 and b are positive constants and positive current is taken to be in the clockwise direction.

- (b) **Derive** an expression for the absolute value of the induced emf in the smaller loop for times $t \geq 0$. Express your answer in terms of r_0 , n_0 , I_0 , b , t , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

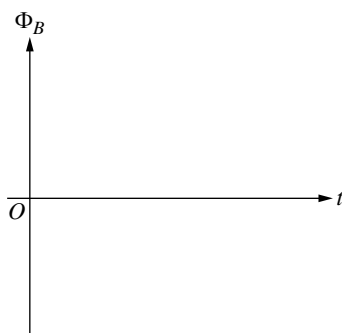


Figure 4

- (c) On the axes in Figure 4, **sketch** a graph of the magnetic flux Φ_B through the small conducting loop due to the current in the solenoid as a function of time t from $t=0$ until the flux reaches zero. On the horizontal axis of the graph, **label** the time at which the flux is zero with an expression in terms of the constant b .
- (d) **Indicate** the factor by which the vertical intercept of the graph you drew in part (c) would change, if at all, if the loop inside the solenoid had a radius of $2r_0$.

Briefly **justify** your answer.

FREE-RESPONSE QUESTION: EXPERIMENTAL DESIGN AND ANALYSIS

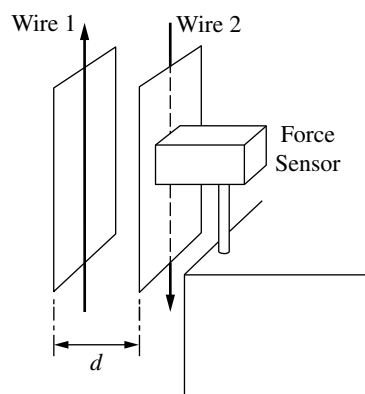


Figure 1

3. Students are investigating the relationship between the distance between current carrying wires and the magnitude of the magnetic force exerted between the wires using the setup shown in Figure 1. Two vertical wires, Wire 1 and Wire 2, are connected to power supplies (not shown) and carry currents in opposite directions. Each wire is attached to a board, and the board with Wire 2 is connected to a stationary force sensor. The boards are held parallel to each other and the distance d between the wires can be adjusted.

The students hypothesize that the force exerted by Wire 1 on Wire 2 is proportional to the inverse of the square of the distance between the wires.

- (a) **Describe** a procedure for collecting data that could be used to determine the relationship between the magnitude of the magnetic force exerted on the wires and the distance between the wires. Include any steps necessary to reduce experimental uncertainty.
- (b) **Describe** what quantities could be graphed and how the graph could be analyzed to test their hypothesis.

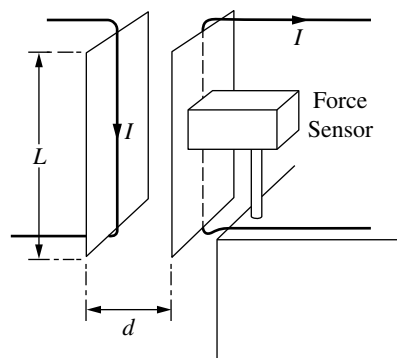


Figure 2

The students want to determine an experimental value for the magnetic permeability of free space, μ_0 . The same experimental setup is used and the current I is the same for both wires, as shown in Figure 2. The length L of the

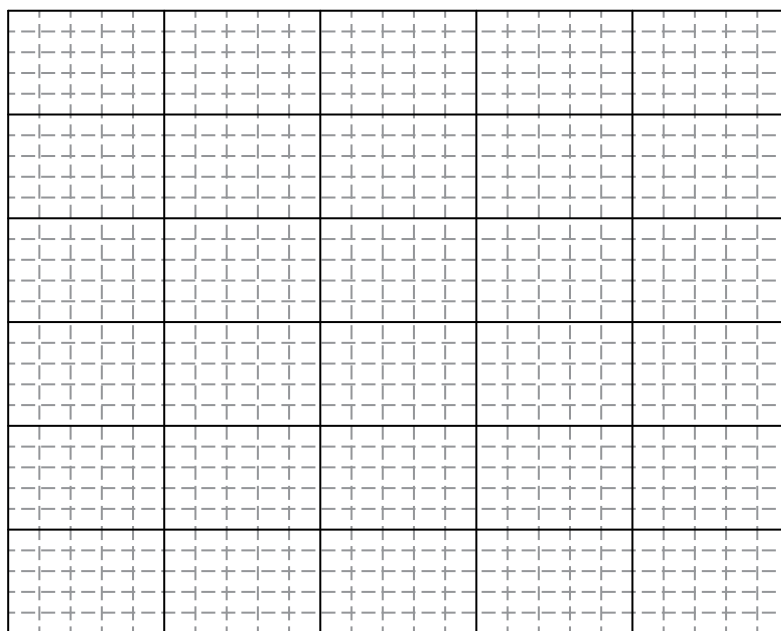
wire segments fixed to the boards are measured. In this experiment the distance d between the wires is held fixed and the force is measured while the current I is then varied. Using measured values of $L = 1.4$ m and $d = 0.005$ m, the students collect the data shown in the table.

I (A)	F (mN)		
2.8	0.35		
3.2	0.60		
3.6	0.65		
4.0	1.00		
4.4	1.10		

- (c) i. **Indicate** what measured and/or calculated quantities could be graphed to yield a straight line that could be used to calculate a numerical value for μ_0 . Use the blank columns in the table to list any calculated quantities you will graph other than the data provided.

Horizontal Axis: _____ Vertical Axis: _____

- ii. **Plot** the data points for the quantities indicated in part (c)(i) on the graph provided. Clearly scale and label all axes, including units, as appropriate.



- iii. **Draw** the best-fit line for the data graphed in part (c)(ii).
- (d) Using the line drawn in part (c)(iii) and the measured values $L = 1.4$ m and $d = 0.005$ m as needed, **calculate** the magnetic permeability of free space, μ_0 .

FREE-RESPONSE QUESTION: QUALITATIVE/QUANTITATIVE TRANSLATION

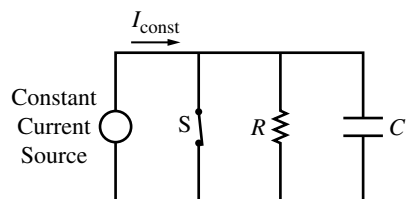


Figure 1

4. The figure shows a circuit consisting of a closed switch S , a resistor of resistance R , and an initially uncharged parallel-plate capacitor of capacitance C in parallel with an ideal constant current source that generates a constant current I_{const} in the direction indicated. At time $t = 0$ the switch S is opened, and the current source continues generating a constant current I_{const} . The potential difference across the capacitor is taken to be positive when it has positive charge on its top plate.

- (a) A long time after the switch has been opened, the rate of change $\frac{d(\Delta V)}{dt}$ of

the potential difference across the capacitor approaches a constant value.

Indicate whether the value is positive, negative, or zero.

___ Positive ___ Negative ___ Zero

Justify your answer.

- (b) **Derive**, but do not solve, a differential equation for the rate of change

$\frac{d(\Delta V)}{dt}$ of the potential difference ΔV across the capacitor after the switch

is opened. Express your answer in terms of R , C , ΔV , t , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

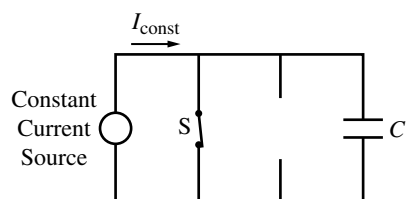


Figure 2

- (c) The process is repeated with the resistor removed from the circuit, leaving a gap in the middle branch, as shown in Figure 2. The switch is again opened with the capacitor initially uncharged, and the current source again generates a constant current I_{const} . **Indicate** whether the rate of change

$\frac{d(\Delta V)}{dt}$ of the potential difference across the capacitor shortly after (but

not immediately after) the switch is opened will be zero, a nonzero constant value, or a value with increasing magnitude.

___ Zero ___ Nonzero constant value ___ Value with increasing magnitude

Justify your answer by referring to the differential equation you wrote for part (b).

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Essential Knowledge
1	D	3.B	13.2.A	13.2.A.1
2	A	2.B	11.4.A	11.4.A.1
3	B	3.B	8.1.A	8.1.A.3
4	C	2.A	12.4.A	12.4.A.1
5	B	2.B	9.2.B	9.2.B.2
6	C	2.A	10.4.A	10.4.A.5
7	D	3.C	13.5.A	13.5.A.3
8	B	3.C	9.3.A	9.3.A.1
9	A	2.A	8.6.A	8.6.A.1
10	D	3.B	11.7.A	11.7.A.2
11	D	2.B	13.2.A	13.2.A.2
12	B	2.A	8.5.A	8.5.A.2
13	A	2.C	12.3.B	12.3.B.1
14	C	2.D	11.5.B	11.5.B.2
15	B	2.C	8.3.A	8.3.A.2

Free-Response Question	Skill	Learning Objective
1	2.A, 3.B, 3.C	8.4.A, 9.2.A, 9.2.B, 9.3.A
2	1.A, 1.C, 2.A, 2.D, 3.B, 3.C,	12.4.A, 13.1.A, 13.2.A
3	1.B, 2.B, 2.D, 3.A	11.3.B, 12.3.B, 12.4.A
4	2.A, 2.D, 3.B, 3.C	10.3.A, 11.1.A, 11.7.A, 11.3.B, 11.8.B



Scoring Guidelines

Scoring Guidelines for Question 1: Mathematical Routines

10 points

Learning Objectives: 8.4.A 9.2.A 9.2.B 9.3.A

(a) i For selecting “Either above or below Point A”, with an attempt at a relevant justification 1 point

For indicating that because the ring and the sphere have opposite sign charges, the net electric potential only depends on the relative distances of the ring and sphere from Point A 1 point

Example Response:

*Either above or below Point A**The net potential at Point A is the sum of the potentials of the ring and the sphere. Because potential is a scalar quantity, and because the ring and the sphere have opposite signs, there are two locations (above and below Point A) where the potential due to the sphere is equal and opposite in sign to the potential of the ring.*(a) ii For a multi-step derivation starting with a statement of conservation of energy (e.g., $E_{\text{total}} = U + K = \text{constant}$) 1 point

For correct expressions for the electric potential energy at two different locations 1 point

For using the correct locations, at both $z = 0$ and $z = -3R$, for the potential energy expressions 1 pointFor substituting correct expressions for the kinetic energy at $z = 0$ and $z = -3R_0$ 1 pointFor a correct answer, $\sqrt{v_o^2 - \frac{2kQ^2}{mR} \left(1 - \frac{1}{\sqrt{10}}\right)}$, or equivalent. 1 point

Example Response:

$$E_{\text{total}} = U + K = \text{constant}$$

$$\left[U + K\right]_{z=0} = \left[U + K\right]_{z=-3R_0}$$

$$-k \frac{Q^2}{R} + \frac{1}{2} m v_o^2 = -\frac{k Q^2}{\sqrt{(3R)^2 + R^2}} + \frac{1}{2} m v^2$$

$$-k \frac{Q^2}{R} + \frac{1}{2} m v_o^2 = -\frac{k Q^2}{\sqrt{10} R^2} + \frac{1}{2} m v^2$$

$$\frac{1}{2} m v^2 = \frac{1}{2} m v_o^2 - k \frac{Q^2}{R} + \frac{k Q^2}{\sqrt{10} R} = \frac{1}{2} m v_o^2 - \frac{k Q^2}{R} \left(1 - \frac{1}{\sqrt{10}}\right)$$

$$v^2 = v_o^2 - \frac{2k Q^2}{m R} \left(1 - \frac{1}{\sqrt{10}}\right)$$

$$v = \sqrt{v_o^2 - \frac{2k Q^2}{m R} \left(1 - \frac{1}{\sqrt{10}}\right)}$$

Total for Part (a)

7 points

(b) For a multi-step derivation starting with an equation for the electric field of a charge distribution **1 point**

For an indication that only the z -component of the field need be considered (e.g., by including a $\cos \theta$ term in the integral) **1 point**

For a correct final answer **1 point**

Scoring note: The answer can be in terms of either ϵ_0 or k .

Example Response

$$\vec{E} = k \int \frac{dq}{r^2} \hat{r}$$

$$E = E_z = k \int \frac{dq}{r^2} \cos \theta = k \int \left(\frac{dq}{(2R)^2 + R^2} \right) \left(\frac{2R_0}{\sqrt{R^2 + 4R^2}} \right)$$

$$E = E_z = k \left(\frac{1}{5R^2} \right) \left(\frac{2}{\sqrt{5}} \right) \int dq$$

$$= k \left(\frac{2Q}{5\sqrt{5}R^2} \right)$$

(b) **Alternate Solution** **1 point**

For a multi-step derivation that relates the electric field to the derivative of the electric potential.

For a correct potential expression as a function of z **1 point**

For correctly differentiating the electric potential expression, substituting $z = 2R_0$ into the derivative, and getting a final answer that is consistent with the electric potential expression found **1 point**

Scoring note: The answer can be in terms of either ϵ_0 or k .

Example Response (Alternate Solution):

$$E = E_z = -\frac{dV}{dz}$$

$$V = k \int \frac{dq}{r} = k \frac{\int dq}{\sqrt{R^2 + z^2}} = k \frac{Q}{\sqrt{R^2 + z^2}} = kQ(R^2 + z^2)^{-1/2}$$

$$\frac{dV}{dz} = kQ \left(-\frac{1}{2} \right) (R^2 + z^2)^{-3/2} (2z) = -kQ \frac{z}{(R^2 + z^2)^{3/2}}$$

$$z = 2R:$$

$$|E| = |E_z| = \left| -\frac{dV}{dz} \right| = kQ \frac{2R}{(R^2 + (2R)^2)^{3/2}} = kQ \frac{2R}{(5R^2)^{3/2}}$$

$$|E| = k \left(\frac{2Q}{5\sqrt{5}R^2} \right)$$

Total for part (b) **3 points**

Total for question 1 **10 points**

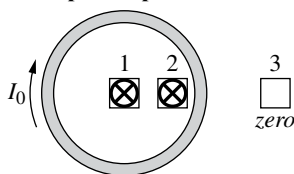
Scoring Guidelines for Question 2: Translation Between Representations

12 points

Learning Objectives: 12.4.A 13.1.A 13.2.A

- | | | |
|-----|---|---------|
| (a) | For indicating consistent fields along the z -axis in Regions 1 and 2 | 1 point |
| | Scoring note: Consistent fields include the following:
A cross within a circle for both regions
A dot within a circle for both regions | |
| | For indicating that for Region 1 the field is into the plane of the figure | 1 point |
| | For writing “zero” next to Region 3 | 1 point |

Example Response:



Total for Part (a) 3 points

- | | | |
|-----|--|---------|
| (b) | For a multistep derivation that starts with $\mathcal{E} = -\frac{d\Phi}{dt}$ | 1 point |
| | For substituting the small loop's area into an expression for the magnetic flux | 1 point |
| | For substituting a correct expression for the magnitude of the magnetic field inside a solenoid | 1 point |
| | For correctly taking the time derivative of an expression for either the magnetic flux, the magnetic field, or the current | 1 point |
- Scoring note:** The expression for flux, field, or current need not be correct to earn this point.

Example Response:

$$A = \pi r_0^2 \text{ and } B = \mu_0 n_0 I$$

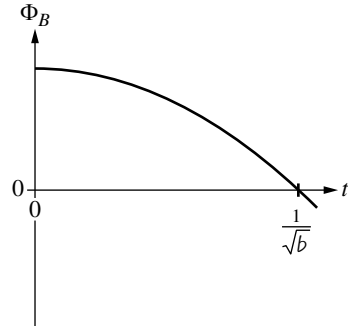
$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

$$|\mathcal{E}| = \left| \frac{d\Phi_B}{dt} \right| = \left| \frac{d(\vec{B} \cdot \vec{A})}{dt} \right| = A \left| \frac{dB}{dt} \right|$$

$$|\mathcal{E}| = (\pi r_0^2) \left(\mu_0 n_0 \left| \frac{dI}{dt} \right| \right) = \pi r_0^2 \mu_0 n_0 |-I_0 (2bt)|$$

$$|\mathcal{E}| = 2bI_0 \pi r_0^2 \mu_0 n_0 t$$

Total for Part (b) 4 points

(c)	For drawing a curve that is concave-downward everywhere	1 point
	For a line or curve that contains the point $\left(\frac{1}{\sqrt{b}}, 0\right)$ on the horizontal axis, with $\frac{1}{\sqrt{b}}$ clearly indicated	1 point
	Scoring note: The curve need not extend to include negative flux values for times $t > \frac{1}{\sqrt{b}}$ to earn this point.	
	For a continuous curve that has a positive maximum on the vertical axis and zero slope at the maximum	1 point
	Scoring note: Drawing the mirror image across the horizontal axis earns full credit.	
	Example Response	
		
	Total for Part (c)	3 points
(d)	For indicating that the intercept will increase, with an attempt at a justification using functional dependence	1 point
	For indicating that the flux (or area) is proportional to the square of the loop's radius so will increase by a factor of 4	1 point
	Example Response	
	<i>The vertical intercept will increase by a factor of 4. This is because the loop area is proportional to r^2, and if the radius doubles, the area and therefore the flux will increase by a factor of 4.</i>	
	Total for part (d)	2 points
	Total for question 2	12 points

Scoring Guidelines for Question 3: Experimental Design and Analysis 10 points

Learning Objectives: 11.3.B 12.3.B 12.4.A

(a) For indicating that current is held constant, distance between the wires is varied, and force is measured 1 point

For indicating that measurements should be taken for several wire distances 1 point

Example Response:

Use the power supply to create constant currents in the wires. Measure the plate separation d . Record the force reading F on the force sensor. Repeat this for a total of 10 values of plate separation distances, using the same wire currents each time.

Total for Part (a) 2 points

(b) For indicating that a graph involving force and distance can be used to determine the relationship 1 point

For describing a valid method of determining whether the graph is consistent with the hypothesis 1 point

Example Response:

Plot F vs $\frac{1}{d^2}$. If it is linear, the hypothesis is correct, if not, it's incorrect.

OR

Plot F vs $\frac{1}{d}$. If it is linear, then the hypothesis is incorrect.

OR

Plot F vs d , use power regression to find functional dependence.

Total for Part (b) 2 points

(c) i For indicating appropriate quantities that would produce a straight-line graph to determine μ_0 1 point

Examples:

Horizontal Axis: I^2

Vertical Axis: F

OR

Horizontal Axis: I

Vertical Axis: $\sqrt{F}$

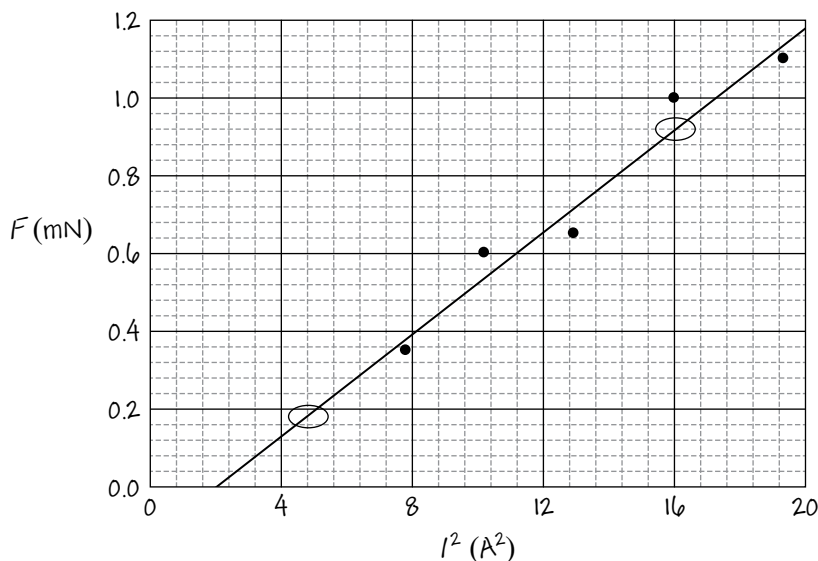
(c) ii For labeling the axes correctly (including units) with a linear scale, where the data uses at least half the range on each axis of the graph 1 point

For plotting the data points correctly 1 point

(c) iii For drawing a straight line that approximates the trend of the data 1 point

Example Response for (c)(i), (c)(ii), (c)(iii) combined:Horizontal Axis: I^2 Vertical Axis: F

$I(\text{A})$	$F(\text{mN})$	$I^2(\text{A}^2)$	
2.8	0.35	7.8	
3.2	0.60	10.2	
3.6	0.65	13.0	
4.0	1.00	16.0	
4.4	1.10	19.4	



Scoring note: The circled points are used for the slope calculation in part (d). It is not necessary to indicate them here to earn credit in part (c).

Total for part (c) 4 points

(d) For using an appropriate feature of the best-fit line to calculate μ_0 1 point

For correctly relating the slope to μ_0 1 point

Scoring note: Linear regression on the given data, graphed as F vs I^2 , yields a slope of 0.0657 mN/A^2 and $\mu_0 = 1.47 \times 10^{-6} \text{ N/A}^2$.

Example Response:

The line contains the points $(4.8 \text{ A}^2, 0.18 \text{ mN})$ and $(16.0 \text{ A}^2, 0.92 \text{ mN})$.

The slope of the line is

$$\text{slope} = \frac{0.92 - 0.18}{16.0 - 4.8} \text{ mN/A}^2 = \frac{0.74}{11.2} \text{ mN/A}^2 = 0.0661 \text{ mN/A}^2 \text{ or } 0.0661 \times 10^{-3} \text{ N/A}^2$$

Force and current are related as follows.

$$F = ILB \quad \text{and} \quad B = \frac{\mu_0 I}{2\pi d}$$

$$F = \frac{\mu_0}{2\pi} I^2 \frac{L}{d}$$

$$\text{So the slope of the graph is } \text{slope} = \frac{\mu_0 L}{2\pi d} \quad \text{and} \quad \mu_0 = 2\pi (\text{slope}) \frac{d}{L}$$

$$\mu_0 = 2\pi (0.0661 \times 10^{-3} \text{ N/A}^2) \frac{0.005 \text{ m}}{1.4 \text{ m}} = 1.48 \times 10^{-6} \text{ N/A}^2$$

Total for part (d) 2 points

Total for question 3 10 points

Scoring Guidelines for Question 4: Qualitative/Quantitative Translation

8 points

Learning Objectives: 10.3.A 11.1.A 11.7.A 11.3.B 11.8.B

- | | | |
|-----|--|---------|
| (a) | For selecting “Zero,” with an attempt at a justification | 1 point |
| | For indicating that the charge builds up to a maximum amount and no longer changes | 1 point |
| | For indicating that a constant charge corresponds to a constant potential difference | 1 point |

Example Response:

Zero.

After the switch is closed, charge will start to build up on the capacitor and it will eventually stop charging. Potential difference is proportional to capacitor charge, so the potential difference will not be changing after a long time.

Total for Part (a) **3 points**

- | | | |
|-----|--|---------|
| (b) | For a multi-step derivation starting with a correct application of either Kirchhoff’s junction rule or loop rule (e.g., $I_{\text{const}} = I_R + I_C$ or $\Delta V_C - \Delta V_R = 0$) | 1 point |
| | For substituting correct expressions of the current in the resistor, $\frac{\Delta V}{R}$, and the current in the capacitor branch, $\frac{dq}{dt}$, into an equation that expresses Kirchhoff’s junction rule | 1 point |
| | For correctly substituting the current in the capacitor branch as $\frac{d(C\Delta V)}{dt}$ | 1 point |

Example Response (Starting with loop rule):

$$\Delta V_C - \Delta V_R = 0$$

$$\Delta V_C = \Delta V_R = I_R R$$

$$I_{\text{const}} = I_R + I_C$$

$$\Delta V_C = (I_{\text{const}} - I_C) R = \Delta V_C = \left(I_{\text{const}} - \frac{dq}{dt} \right) R$$

$$\Delta V_C = \left(I_{\text{const}} - \frac{d(C\Delta V)}{dt} \right) R$$

$$\frac{d(\Delta V)}{dt} = \frac{I_{\text{const}}}{C} - \frac{\Delta V_C}{RC}$$

Example response (Starting with junction rule):

$$I_{\text{const}} = I_R + I_C = \frac{\Delta V}{R} + \frac{dq}{dt}$$

$$I_{\text{const}} = \frac{\Delta V}{R} + \frac{d(C\Delta V)}{dt}$$

$$C \frac{d(\Delta V)}{dt} = I_{\text{const}} - \frac{\Delta V}{R}$$

$$\frac{d(\Delta V)}{dt} = \frac{I_{\text{const}}}{C} - \frac{\Delta V_C}{RC}$$

Total for Part (b) **3 points**

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- (c) For attempting to use functional dependence in the differential equation in part (b) by relating resistance to either a relevant term in the equation or to the derivative of the potential difference across the capacitor **1 point**
- Scoring note:** To earn this point it is not necessary to indicate that the resistor is removed or the resistance becomes infinite, and it is not necessary to use the functional dependence correctly.

For a selection and an explanation that are both consistent with the equation from part (b), and an explanation in which the resistance is infinite or approaches infinity **1 point**

Example Response:

Nonzero constant value.

If the resistor were not present, that is the same as letting R go to infinity. In the differential equation,

this means that the term $\frac{\Delta V}{R}$ goes to zero and can be ignored. The result is $\frac{d(\Delta V)}{dt} = \frac{I_0}{C}$, which is a positive constant.

Total for part (c) **2 points**

Total for question 4 **8 points**